

A Novel Redirected Walking Algorithm for VR Navigation in Small Tracking Area

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ABSTRACT

In VR, how to map large-scale virtual space to small real space is a challenging problem. We propose a novel steering algorithm for the redirected walking(RDW) technique to direct users away from the boundary of the tracking area. When the user arbitrarily specifies the start and end points of the roaming, our system will generate a navigation path based on Voronoi automatically. During the navigation, we interactively and imperceptibly rotate the VE to make the user walking along arcs while thinking she is walking straightly. When the user is approaching the boundary, we hierarchically adjust the redirection gains to make the user turn away to avoid the reset procedure. To validate our algorithm, we conduct live-user comparison experiments with S2C method in a small tracking area using indoor(highly-constructed Maze) and open(long-straight Road) scenes which can be up to three orders of magnitude larger than the tracking area. The results indicate that our algorithm can effectively speed up and smooth the navigation, reduce collisions and perceptual distortion, show the potential to direct multiple users simultaneously, and be particularly suitable for individuals and small business users who cannot provide a large physical space.

Index Terms: Human-centered computing—Human computer interaction(HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

With the advent of more affordable portable devices, interest in immersive virtual reality (VR) has grown rapidly. Locomotion is one of the most important forms of interaction in VR, there are a variety of locomotion techniques, such as the flying, leaning, treadmills, walking-in-place, etc. Among them, in terms of presence, ease of use, and naturalness, *real walking* that allows users with head-mounted displays(HMDs) to navigate through in a virtual environments(VE) is a better method.

In real walking, the user's movement in virtual space is the same as her movement in physical space. One serious problem with real walking is that the tracking area must be as large as the virtual space, otherwise the fidelity of spatial perceptual feedback to the users will be reduced. This limitation makes real walking infeasible for many virtual scenes and facilities. To address this problem, Razzaque proposed a technique of *redirected walking*(RDW) to explore large VEs [2]. RDW works by making the user turn herself by interactively and imperceptibly rotating the VE around her. Theoretically, if the tracking area is large enough, the technology could cause the user to walk in circles in real space without knowing it, thinking she is walking in an endless straight line in VE.

However, it is difficult to leverage unnoticeable perceptual manipulations like imperceptibly small visual translations and rotations to

redirect the user away from the boundary of the tracking area in a small physical space. In this paper, we are committed to designing a novel steering algorithm for RDW to guide individuals or small business users exploring large-scale VEs in a small tracking area without running into the boundary. For a given large-scale VE with its 2D map, after the user specified the start and end points of the navigation, our algorithm will automatically generate a virtual navigating path (termed as *VPath*) based on the Voronoi skeleton. When the user starts to walk in the tracking area, we interactively and imperceptibly rotating the VE to make the user walking along arcs while thinking she is walking straight; if a collision is unavoidable, we hierarchically adjust the redirection gains to make the user turn away within a buffer zone to avoid the reset procedure.

To validate the effectiveness of our algorithm, we conducted a pilot live-user study. Participants are required to complete exploring tasks in Maze and long-straight Road virtual environments, in which the VEs can be one, two, or even three orders of magnitude larger than the tracking area. Experimental results show that compared with the S2C method, our algorithm can effectively and smoothly navigate users walking through large-scale virtual space in a small tracking area, and succeed in reducing collisions and perceptual distortion.

2 ALGORITHM

A good RDW steering algorithm should be able to reduce the number of collisions and reduce perceptual distortion as much as possible. When the tracking area is too small, such as less than 3 square meters, it is difficult to achieve a good redirection effect.

In this algorithm, we introduce a concept of a *virtual wall* which is built inside the physical wall(the boundary of the tracking area), and the area between the physical and virtual walls is the *buffer zone*. When the user is approaching the buffer zone, we adjust the redirection gains hierarchically so that the user can turn around in the buffer zone to avoid the traditional reset procedure.

The main idea is as follows: for a given VE with its 2D map, the user can arbitrarily specify two points as the start and end points of roaming. For the specified two points, our algorithm will generate a virtual navigating path (*VPath*) based on the Voronoi skeleton automatically; see Figure 1 for an example. During the navigation, all straight-line segments of the *VPath* are arced with a minimal radius of 7.5m as used in [1]. When the user reaches the boundary of the buffer zone, we adjust the redirection gains hierarchically, so that the user can complete the turn away from the boundary during walking, and always stays in the buffer zone.

When the user approaches the buffer zone, if the user's subsequent behavior is to turn right or left, just let the user turn, otherwise, the algorithm will direct the user to do the following two procedures:

Procedure 1: Turning around (half-circle, with a radius of R_s). When the user approaches the buffer zone, if the user's subsequent behavior is to continue walking along the current path, the algorithm will guide the user to walk counterclockwise(or clockwise, depending on the user's orientation) a half circle with a radius of R_s .

Procedure 2: Walking into the interior (a quarter circle, with a radius of R_m). When the user stays in the buffer zone, we guide the

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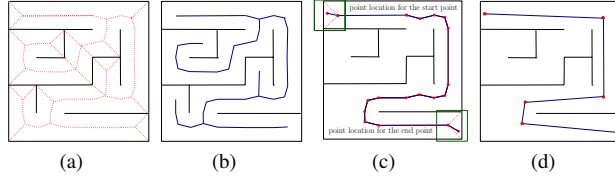


Figure 1: (a) Voronoi diagram of the polygon. (b) Voronoi skeleton. (c) Skeleton path with specified start and end points. (d) The simplified result of the path in (c).

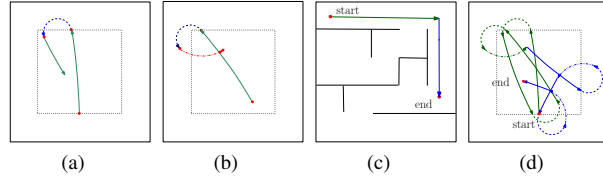


Figure 2: The arrows indicates the walking direction. When the user approaches the buffer zone, two procedures can be invoked. (a) Turning around half a circle with radius R_s (blue dashed arc). (b) Walking into the interior along a quarter circle with radius R_s (red dash-dotted arc). (c) VPath consists of two straight-line segments in the virtual space. (d) RPath consists of arcs in the real space.

user to walk a quarter of a circle with a radius of R_m and let her leave the buffer zone quickly(see Figure 2(b)).

The example in Figure 2(c) and Figure 2(d) show the whole process of navigation.

3 PILOT USER STUDY

Hardware Setup A cardboard HMD composed of cardboard and a smartphone and Kinect are needed to navigate the virtual scene(see Figure 3(a)). During the walking, the gyroscope sensor in the smartphone tracks the user's rotations, and the Kinect tracks the user's position.

Virtual Environment We set up two different virtual environments VE1 and VE2(see Figure 3(b)). VE1 is a highly-structured Maze scene that can be configured in three different sizes: Small($10m \times 10m$), Medium($50m \times 50m$) and Large($100m \times 100m$). VE2 is an open scene, with a long-straight Road with varied lengths($100m$, $200m$, and $500m$, respectively). The tracking area is a $2.5m \times 2.5m$ rectangular region.

Parameters The curvature radius is set to $7.5m$ as suggested in [1]. Because the size of the tracking area is small, we set the width of the buffer zone R_s and R_m as $0.5m$ and $0.25m$, respectively.

Participants We conduct a user study with 15 participants(8 female, mean age 24.56 ± 3.56).

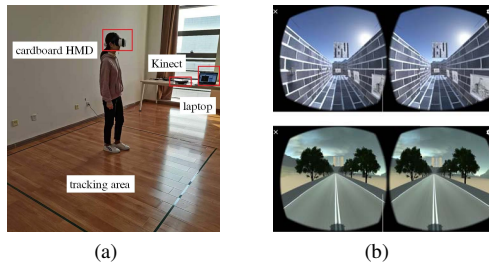


Figure 3: (a) Users HMD view. (b) Hardware setup.

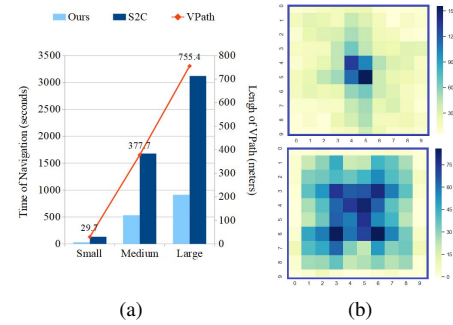


Figure 4: (a) Comparison of navigation time between the S2C and our algorithms for 3 tasks in VE1. (b) An example of the spatial distribution by using the S2C (top) and our algorithm (bottom) when steering the user in a Large Maze.

Time of Navigation. In S2C, a reset operation is activated when the user touches any boundary of the tracking area. According to our statistics, the average time for doing one is approximately 8 seconds. In our algorithm, no reset is needed at all. Figure 4 shows the comparison of navigation time between the S2C and our algorithms under the 3 walking tasks in Maze environments, which indicates that our algorithm shows a significant advantage in improving the time of navigation in the VE1 configuration.

Redirection Rate. In the S2C method, the user spent more than 75% time to reset. While, by using our method, the user only spends less than 32% time to walk along tight arcs and feel the rotation of the VE slightly. In conclusion, using our method, the redirect rate may increase, but the user can explore VEs faster and smoother.

Spatial Distribution. The statistics in Figure 4(b) show that participants' footprints are all over the tracking area when using our algorithm. So our algorithm has more advantages in handling simultaneous navigation of multiple users.

4 CONCLUSION

In this paper, we propose a novel steering algorithm for the RDW to guide the user to explore a large-scale virtual space in a small-scale tracking area. In our algorithm, after the user specifies the start and end points of the navigation, the system will generate a navigation path based on the Voronoi skeleton path. By introducing the curvature gains, we interactively and imperceptibly rotating the VE to make the user walking along arcs in real space while thinking she is walking straightly in the virtual space. When the user approaches the boundary, we hierarchically adjust the redirection gains to make the user turn around in the buffer zone to avoid the reset procedure. Experimental results show that our algorithm can effectively speed up and smooth the navigation, reduce collisions and perceptual distortion dramatically, and show the potential to guide multiple users at the same time.

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REFERENCES

- [1] E. Hodgson, E. Bachmann, and D. Waller. Steering immersed users of virtual environments: Assessing the potential for spatial interference. *ACM: Transactions on Applied Perception*, 8:1–22, 01 2011.
- [2] S. Razzaque, Z. Kohn, and M. C. Whitton. Redirected Walking. In *Eurographics 2001 - Short Presentations*. Eurographics Association, 2001. doi: 10.2312/egs.20011036